

Ask for JSON: Structured Output Collapses Answer Diversity Across 44 Language Models

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Abstract

When a language model must choose one answer from a large space of equally valid options, a format clause — “*Reply with JSON only*” — changes which answer it chooses. We re-run the One-Word Census [1], 31 category prompts with wide answer spaces asked of 44 models, with the reply requested in JSON: no schema enforcement, no constrained decoding, only the request. The field’s convergence deepens sharply: on the unconstrained prompt (“Pick a word.”), the modal answer’s share rises from 40% to 63% and distinct answers fall from 53 to 28; mean answer-choice surprisal drops 1.80 to 1.58 bits. The tax is *progressive* and respects the instrument’s resolution: seven of 44 models move individually (BH-FDR $q = .10$), all toward the mode, led by the panel’s two most distinctive frontier models (the strongest explorer loses 1.32 bits — half its measured distinctiveness) while the conformist floor is unmoved; one family-level effect survives aggregation — the Llama-derived models (excepting a heavily third-party-retuned one) *diverge* under JSON (+0.70 bits against the rest of the field, $p < 10^{-4}$) — and the remaining deltas form a noise plateau we decline to rank. The format acts as a sharpener, not a re-indexer — the plain-chat modal answer survives in 28 of 31 categories, and the three flips occur exactly where the plain mode was weakest. The column also shows that a model’s *defaults are register-indexed*, read from discrete behavior rather than score deltas: of the panel’s 144 stable off-modal defaults (the same non-modal answer, four runs of four, in chat), only 47% survive into JSON. Retention is a per-model trait — GPT-5.6 Terra loses its signature defaults (*mustard*→*ketchup*, *mango*→*apple*, four-of-four flips), while Claude Fable 5 retains most of its own (5 of 7) and acquires *new* stable defaults that exist only in the register (*cerulean*, *carpenter*, four-of-four in JSON, never in chat). Personality is indexed to the channel, as the census found answers indexed to wording. Full-battery controls reveal a register gradient: the compression is significant and specific to the answer-delivery formats models are trained to speak (JSON -0.22 bits, $p = 10^{-4}$; XML -0.18 , $p = .003$; sign-flip permutation over models, corroborated over categories), absent for YAML and CSV ($p > .15$), and *reversed* for an arbitrary non-data wrapper — square brackets significantly loosen the field (+0.16, $p = .002$). On the unconstrained prompt, all four serializations concentrate the pool (serendipity 41% plain → 52–65%), so a corpus-register component remains visible; but the battery-wide gradient weighs toward tool-use post-training as the dominant mechanism. Structured output is how software consumes language models; that surface is served by a measurably more homogeneous model than the chat surface on which models are evaluated, compared, and chosen.

1 Introduction

2 Related work

3 Method

3.1 Instrument

3.2 Format columns

3.3 Parsing and hygiene

4 Results

4.1 The format tax is progressive

4.2 A sharpener, not a re-indexer

4.3 The register moves the mode, not the temperature

4.4 Conviction vs. costume

4.5 Serialization vs. structure: the register gradient

5 Limitations

6 Discussion

Data and code availability

Note on AI usage

References

- [1] Tapan Parikh. The one-word census: Answer-choice conformity across 39 language models, 2026. arXiv preprint; data and explorer at <https://github.com/tap2k/modelun>.